SOP - BusinessLogicAutomationAndWebhooks - Parent

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Document ID:     SOP - BusinessLogicAutomationAndWebhooks - Parent
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        SOP (Standard Operating Procedure)
Purpose:         BusinessLogicAutomationAndWebhooks
Tier / Level:    Parent (Master Specification)
Status:          Locked 🔒 (Master Baseline)
Target Audience: Lead TechOps Specialists, Automation Engineers, Systems Coordinators
Ecosystem Scope: Shopify Flow, Shopify Webhooks API, Gorgias, 3PL WMS (ShipBob), Klaviyo
Business Unit:   Synapse Botanicals (Automation & Systems Infrastructure)
```

1. Purpose, Scope & Automation Governance

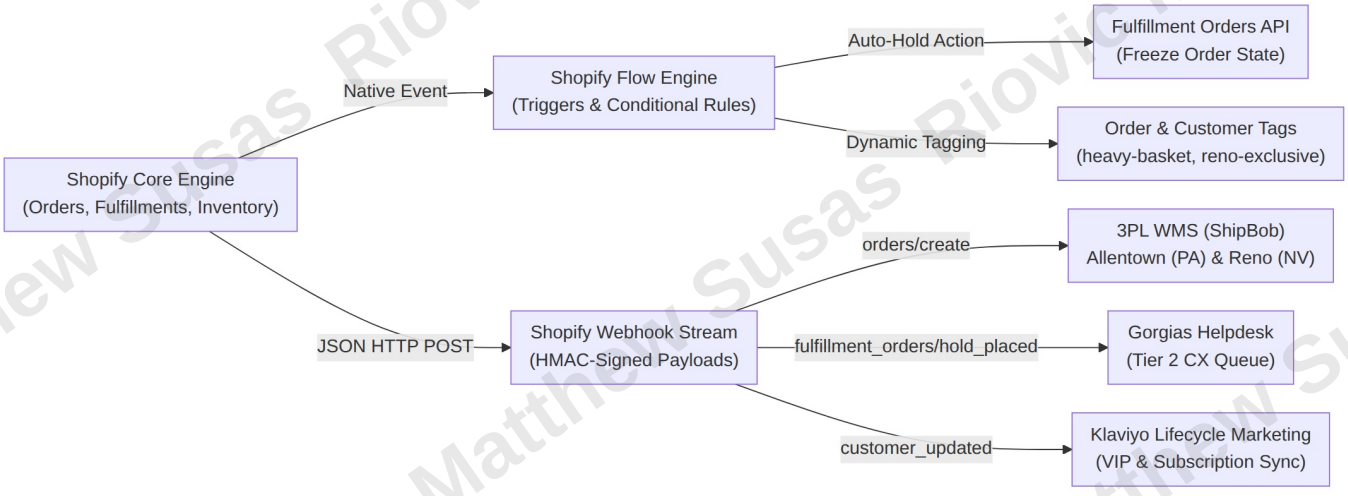
1.1 Purpose

This Standard Operating Procedure (SOP) defines the mandatory engineering standards for constructing, deploying, monitoring, and debugging business logic automations and webhook integrations across Synapse Botanicals. Strict adherence to this protocol prevents:

- **Infinite Tag Recursion Loops:** Runaway automations that continuously mutate order and customer tags, depleting Shopify API rate limits and flooding audit logs.
- **Silent Webhook Dropouts:** Unmonitored 3PL endpoints that exceed failure thresholds, causing Shopify to permanently disable data transmission without notifying customer support.
- **Asynchronous State Drift:** Automations executing actions based on stale order data following a time delay (e.g., releasing a hold on an order that was canceled while waiting).
- **Security & Payload Tampering:** Unverified webhook endpoints processing spoofed or malicious HTTP requests.

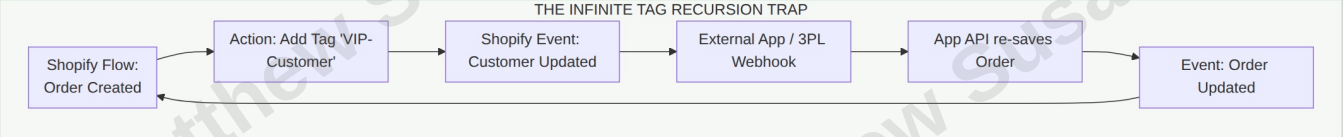
1.2 System Architecture & Event Flow

The Synapse Botanicals event-driven ecosystem connects customer actions on Shopify to distributed operations:



2. The Two Fatal Automation Traps & Architectural Safeguards

2.1 Trap A: The Infinite Flow Tag Recursion Loop



The Mandatory Safeguard: The Idempotency & Negative Check Rule

Every Shopify Flow that adds a tag, updates a metafield, or places a hold **must implement a strict negative condition** checking that the target state does *not* already exist:

- **Rule:** Never execute an `Add order tags` action without first validating: `Order tags` → `does not include` → `[TARGET_TAG]`
- **Rule:** Never execute a customer tag mutation from an order trigger without checking customer tags first.

> **CAUTION THE RUNAWAY AUTOMATION CASCADE**
> In Shopify's event model, adding a tag to a customer or order triggers an `orders/updated` or `customers/updated` event. If a second automation or third-party app listens to that update and modifies the record, an infinite loop is born. Within 15 minutes, a runaway loop can:
>
> 1. Consume 100% of your store's Shopify API rate limit bucket (Leaky Bucket algorithm: 40 requests/sec).
> 2. Lock up checkout and customer account portal response times.
> 3. Generate over 50,000 spam entries in the Shopify Order Timeline.

2.2 Trap B: Silent Webhook Dropouts & The 19-Failure Rule

The Mandatory Safeguard:

1. Webhook endpoints must immediately return an HTTP `200 OK` acknowledgment upon packet arrival, delegating heavy database processing to an asynchronous internal queue.
2. TechOps must execute a bi-weekly webhook endpoint health check under **Settings → Notifications → Webhooks**.

> **WARNING SHOPIFY'S AUTOMATIC WEBHOOK TERMINATION POLICY**
> When an external endpoint (e.g., ShipBob, Gorgias, or an internal analytics server) returns an HTTP error (`500 Server Error`, `504 Gateway Timeout`) or takes > 5 seconds to respond:
>
> - Shopify retries delivery up to **19 times** over a 48-hour period with exponential backoff.
> - If all 19 attempts fail, **Shopify silently disables the webhook subscription entirely.**
> - No notification is sent to customer service. The 3PL simply stops receiving new orders, leading to undetected warehouse fulfillment halts that can persist for days.

3. SOP - BusinessLogicAutomationAndWebhooks - Protocol A: Shopify Flow Architecture & Standard Design Patterns

Shopify Flow automations in Synapse Botanicals must follow strict structural guidelines to maintain clarity, prevent race conditions, and guarantee predictable execution.

3.1 Standard Trigger → Condition → Action Design Schema

```
┌──────────────────────────────────────────────────────────────────┐
│                        FLOW DESIGN HIERARCHY                     │
├──────────────────────────────────────────────────────────────────┤
│  1. TRIGGER:     Specific system event (e.g., Order created)     │
│  2. GATE 1:      Safety / Negative Check (Does target tag NOT exist?)  │
│  3. GATE 2:      Business Criteria (Weight, SKU, Fraud recommendation)  │
│  4. WAIT DELAY:  Optional buffer window for customer self-service edits│
│  5. RE-GATE:     Re-verify order state post-delay                │
│  6. ACTION:      Execute mutation (Fulfillment Hold, Add Tag)    │
└──────────────────────────────────────────────────────────────────┘
```

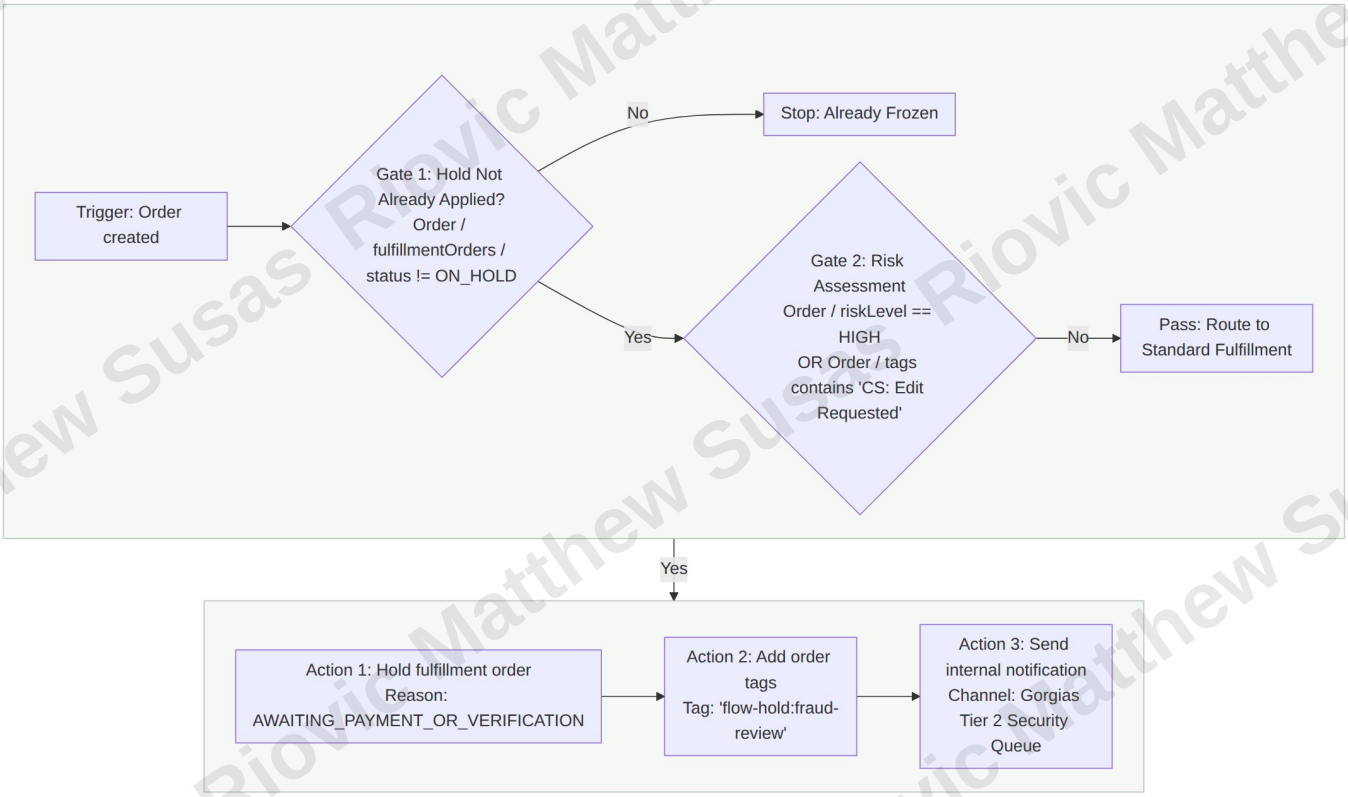
3.2 Managing Asynchronous Wait Delays & State Drift

When utilizing Shopify Flow's native **Wait** action (e.g., delaying fulfillment for 15 minutes to allow customer address corrections):

1. **Never assume the order is in the same state post-delay.**
2. A customer may have canceled their order, or Tier 2 CX may have already applied a manual hold during the 15-minute window.
3. **Mandatory Post-Delay Re-Check:** Always place a condition directly after the `Wait` node verifying: `Order / fulfillmentStatus` is equal to `UNFULFILLED` **AND** `Order / cancelledAt` is `null`.

3.3 Production Blueprint 1: High-Risk Order Fraud & Hold Automation

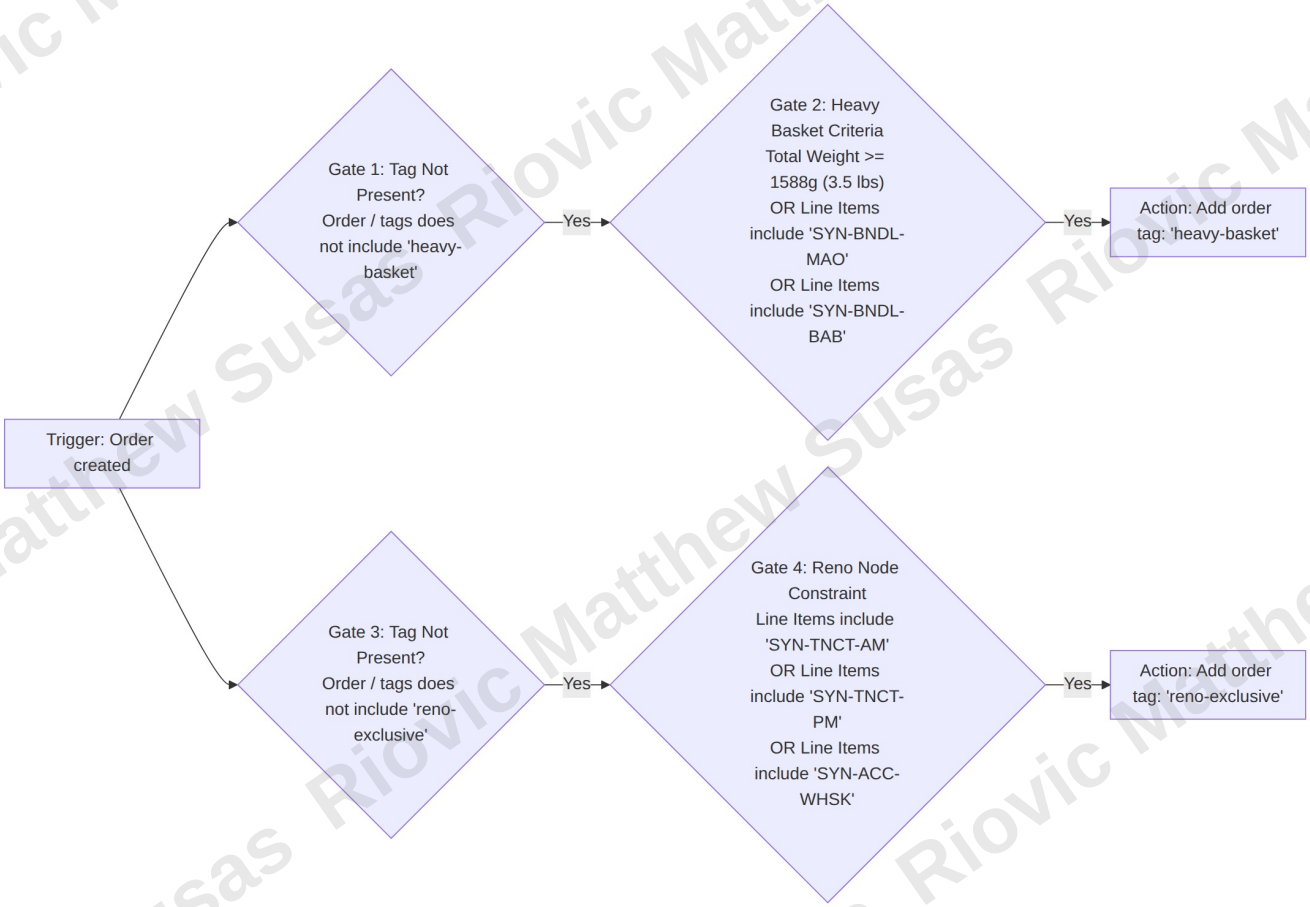
This automation protects store revenue by automatically intercepting suspicious orders before warehouse distribution centers can pick or pack them.



> **NOTE System Architecture Dependency:**
> The fulfillment hold reason and manual release gates triggered by Action 1 operate under the strict operational protocols defined in **SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)** – **Section 2.2** and the state machine in **Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)** – **Section 1.1**.

3.4 Production Blueprint 2: Heavy-Basket & Location Tagging Automation

This flow inspects new orders to dynamically assign operational tags that govern multi-node warehouse allocation.



> **NOTE System Architecture Dependency:**
> The `heavy-basket` tag triggers Rule 2 consolidated shipping directives, while `reno-exclusive` triggers Rule 1 location restrictions in **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)** – **Section 3.2** based on the 3.5 lb breakeven curve modeled in **Matrix - OrderRoutingCostOptimization - DecisionEngine (Domain 2.2)** – **Section 2.2**.

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4. SOP - BusinessLogicAutomationAndWebhooks - Protocol B: Loop Circuit-Breakers & Tag Hygiene

To prevent app conflicts and runaway tag loops, all tags created by automated systems must follow the Synapse Botanicals **Tag Namespace Taxonomy**.

4.1 Standardized Tag Namespace Taxonomy

Tag Prefix	Intended Creator	Purpose & Downstream Consumer	Example Tag	Deletion Rule
`flow-hold:`	Shopify Flow	Identifies active automated fulfillment holds for Tier 2 CX	`flow-hold:fraud-review`, `flow-hold:address-verify`	Removed automatically upon hold release
`flow-route:`	Shopify Flow	Signals order routing engine logic directives	`flow-route:heavy-basket`, `flow-route:reno-only`	Permanent audit record; do not delete
`sys:`	3PL WMS / Integrations	Confirms external warehouse sync state	`sys:shipbob-allocated`, `sys:shipbob-allocated`	Read-only; managed by webhook payload
`cs:`	Gorgias / Tier 2 CX	Manual operator intervention tags	`cs:manual-audit-cleared`, `cs:address-override`	Managed by customer experience staff

4.2 The 3-Point Circuit-Breaker Rule

Before activating any Shopify Flow that mutates data, the author must confirm all three circuit breakers are present:

1. **Negative Existence Check:** Condition explicitly checks that the action has not already run on this object.
2. **Namespace Isolation:** The automation only triggers on or modifies tags within its designated prefix (e.g., `flow-route:`), never modifying generic un-prefixed tags.
3. **Execution Cap:** If an automation loops or triggers more than 5 times on the same object ID within 60 seconds, Shopify Flow engine throttling terminates the run automatically.

> **NOTE Data Governance Reference:**
> Tag formatting rules, kebab-case syntax, and non-colliding namespace constraints are enforced under **Matrix - CatalogDataSchemaAndValidation - Governance (Domain 3.2)** – **Section 3 (Field 6)**.

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5. SOP - BusinessLogicAutomationAndWebhooks - Protocol C: Webhook Endpoint Configuration & HMAC Security

Webhooks broadcast real-time operational events from Shopify to external cloud services. Every webhook subscription must be rigorously configured and cryptographically authenticated.

5.1 Mandatory Webhook Subscriptions for Synapse Botanicals

Webhook Event Topic	Primary Receiving Endpoint	Format	API Version	Operational Purpose
`orders/create`	3PL WMS (ShipBob)	JSON	`2026-10`	Ingests new orders into warehouse picking queues
`orders/updated`	3PL WMS & Gorgias	JSON	`2026-10`	Synchronizes address corrections and variant line-item edits
`fulfillment_orders/hold_placed`	Gorgias & Slack Alerts	JSON	`2026-10`	Alerts CX team that an order is frozen downstream
`inventory_levels/connect`	Master ERP Ledger	JSON	`2026-10`	Tracks multi-location inventory provisioning at PA and NV nodes

5.2 Cryptographic HMAC Verification Standard

To prevent attacker-spoofed webhook payloads from injecting fake orders or altering inventory counts, every endpoint receiving Shopify webhooks must execute cryptographic SHA-256 HMAC verification before parsing the request body.

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                        HMAC DIGITAL SIGNATURE VALIDATION FLOW                │
├──────────────────────────────────────────────────────────────────────────────┤
│ 1. Extract raw byte body from incoming HTTP POST request.                    │
│ 2. Compute HMAC-SHA256 hash using the shared store secret key.               │
│ 3. Base64-encode the resulting hash.                                         │
│ 4. Compare calculated hash to header: X-Shopify-Hmac-Sha256.                 │
│ 5. If MATCH: Return HTTP 200 OK and queue task. If MISMATCH: Return HTTP 401 & abort. │
└──────────────────────────────────────────────────────────────────────────────┘
```

Production Verification Code (Node.js Reference):

```
        const crypto = require('crypto');

        function verifyShopifyWebhook(rawBody, hmacHeader, secretKey) {
          const calculatedHmac = crypto
            .createHmac('sha256', secretKey)
            .update(rawBody, 'utf8')
            .digest('base64');

          // Use timing-safe comparison to prevent side-channel timing attacks
          return crypto.timingSafeEqual(
            Buffer.from(calculatedHmac),
            Buffer.from(hmacHeader)
          );
        }
```

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6. SOP - BusinessLogicAutomationAndWebhooks - Protocol D: Triage, Run-Log Diagnostics & Emergency Kill-Switches

When an automation or integration fails, TechOps must isolate and resolve the issue within standard Service Level Agreements (< 15 minutes).

6.1 Diagnosing Shopify Flow Execution Run Logs

To investigate why an automated action failed or why a condition evaluated unexpectedly:

1. In Shopify Admin, navigate to **Apps → Shopify Flow**.
2. Click on the target workflow → Select the **Run History** tab.
3. Locate the target Order ID or timestamp.

Run Log Status	Technical Meaning	Root Cause Diagnosis	Immediate Remediation Action
`Completed`	All actions executed successfully	Normal execution	None required
`Action failed`	Trigger and conditions passed, but an action threw an API error	API field constraint violated (e.g., trying to place a hold on an already fulfilled order)	Review action payload; verify order state prior to action
`Waiting`	Workflow is currently paused inside a `Wait` node	Asynchronous delay timer is running	Normal; check scheduled resume timestamp
`Timed out`	Action or external call took > 30 seconds to respond	Downstream endpoint latency or GraphQL query timeout	Optimize GraphQL payload; verify app server responsiveness

6.2 Webhook Delivery Audit & HTTP Error Triage

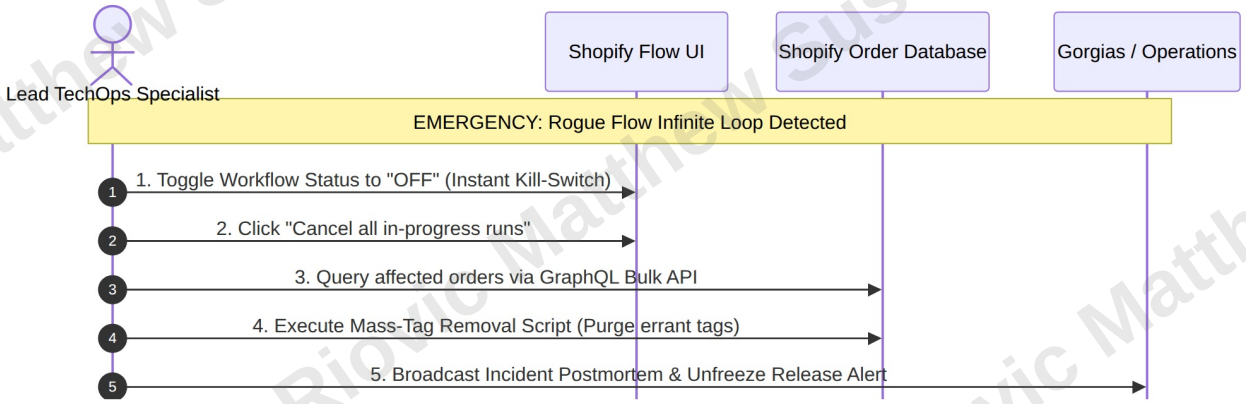
To inspect failing webhook event deliveries:

1. Navigate to **Shopify Admin → Settings → Notifications → Webhooks**.
2. Locate endpoints displaying warning indicators or consecutive failure counts.

HTTP Status Code	Meaning	Root Cause	Triage & Remediation Protocol
`401 Unauthorized`	Authentication Failure	Shared secret key mismatch or expired app token	Re-generate API secret in receiving app; update webhook subscription
`422 Unprocessable`	Schema Rejection	Missing mandatory field in payload or invalid format	Verify payload matches endpoint schema (e.g., missing address postal code)
`500 Server Error`	Internal Server Crash	Unhandled exception in receiving app's database handler	Inspect 3PL server error logs; fix unhandled null values
`504 Gateway Timeout`	Response Window Exceeded	Receiving server took > 5 seconds to respond	Refactor endpoint to immediately acknowledge HTTP 200 before background processing

6.3 Emergency Automation Kill-Switch Protocol

If an active flow triggers an infinite loop, spams customer tags, or errantly freezes all store orders:



1. **Step 1: Deactivate the Workflow Immediately:**
   Open the workflow inside **Shopify Flow** → Click **Turn off workflow** in the upper right corner.
2. **Step 2: Terminate In-Progress Delayed Queues:**
   Click **Cancel scheduled runs** to purge all orders currently sitting in `Wait` nodes, preventing delayed actions from firing.
3. **Step 3: Execute Tag Remediation:**
   Use Matrixify to export orders tagged within the incident window → Clear errant tags → Re-import via `MERGE`.

> **NOTE Audit Trail Reference:**
> For the companion error response triage engine and circuit-breaker state diagrams, consult **Matrix - AutomationLogicAndWebhookTriage - Engine (Domain 4.2)**.

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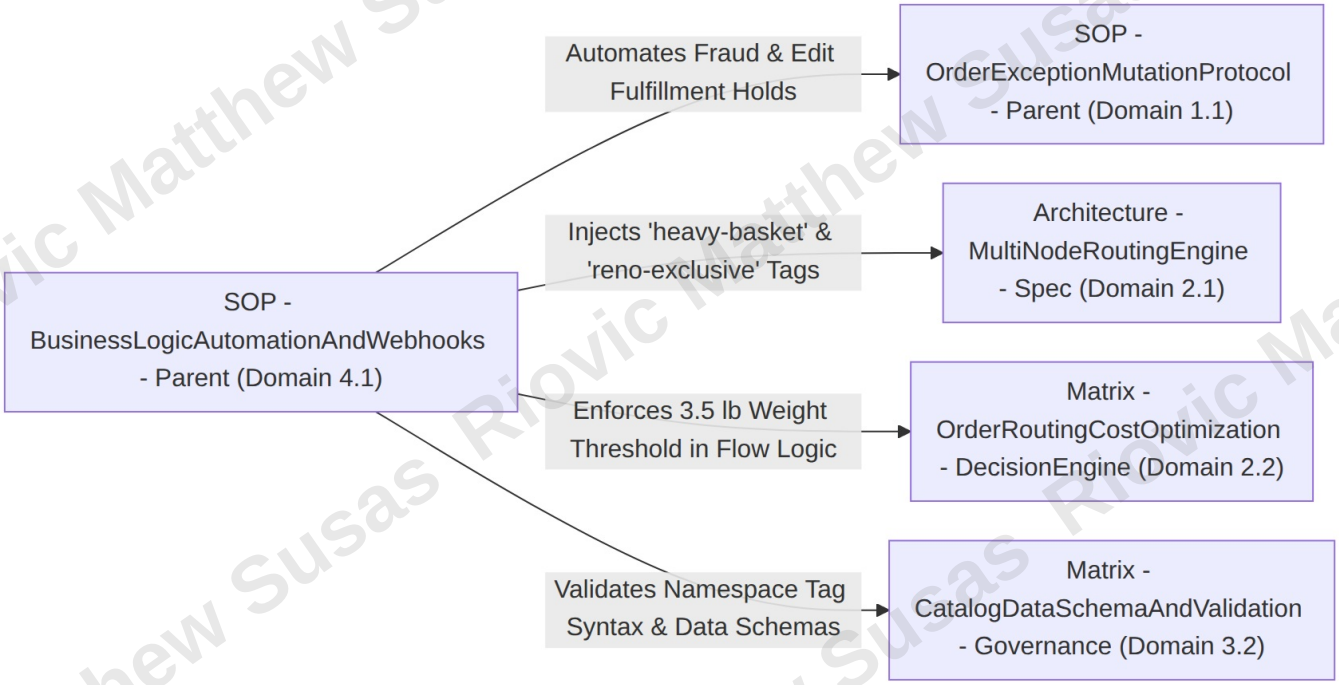
7. Master Domain Interconnectivity Matrix

This Master SOP establishes foundational event-driven connectivity across all Synapse Botanicals technical operational suites:



- **SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1):**
  - **Section 2.2 & 6.1:** Automated hold placements defined in Blueprint 1 (Section 3.3) trigger the formal order exception holding lifecycle.
- **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1):**
  - **Section 3.2:** Dynamic basket tags (`heavy-basket`, `reno-exclusive`) injected by Blueprint 2 (Section 3.4) govern Rule 1 and Rule 2 routing evaluations.
- **Matrix - OrderRoutingCostOptimization - DecisionEngine (Domain 2.2):**
  - **Section 2.2:** Supplies the empirical 3.5 lb weight threshold applied in the Flow condition gate.
- **Matrix - CatalogDataSchemaAndValidation - Governance (Domain 3.2):**
  - **Section 3 (Field 6) & Section 4:** Regulates the tag formatting standards and custom metafield structures updated by Flow actions.